

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8375**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Sect. No:	2022600001
Course Section:	1				

L_61_459_986_L_1505_48213



Name of the candidate. **AAVUSHMAN DASH**

Candidate's UID number:

2	0	2	2	6	0	0	0	0	1
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Examination class room number:

2	0	2
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Invigilator's signature with date:

 08/10/2025

(To be filled in by the candidate)

EXAMINATION: **BE B.TECH MSE**
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: **AIML**

I / II / III / IV / V / **VI** / VII / VIII

DAY AND DATE: **WEDNESDAY, 8/10/25**

COURSE NAME: **DL**

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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$$a) a) f(w) = \frac{1}{2} \|xw - y\|^2$$

(Q1) A)

$$\frac{\partial f}{\partial w} = \frac{1}{2} \cdot 2 \|xw - y\| \cdot (x) = \|xw - y\| x$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot 2 \|xw - y\| (-1) = -\|xw - y\|$$

$$= \frac{\partial f}{\partial w} + \frac{\partial f}{\partial y} = \|xw - y\| (x - 1)$$

$$h_n = h_0 + \lambda Q$$

$$\frac{\partial L_n}{\partial h} = \frac{\partial L}{\partial h} + \lambda \frac{\partial Q}{\partial h}$$

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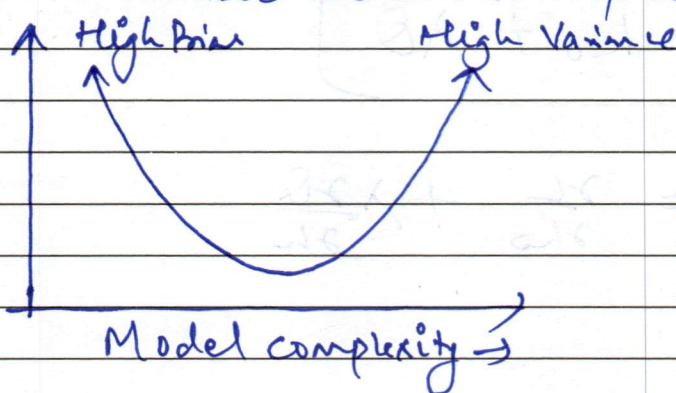
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Q1) b) Bias Variance Decomposition

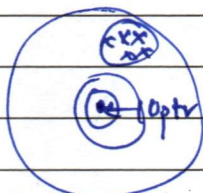
Q1) b)

Bias: It is the error that occurs when the model is not able to capture patterns, relationships, complexities etc from the training data leading to underfitting.

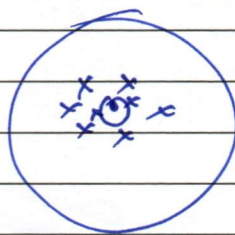
Variance: It is error that occurs when a given model even tries to capture the noise & disturbances of a given data leading to the condition of overfitting.



Bias can be understood using an example like an archer tries to shoot but all his arrows are far off from 10 ptr. Even though it would have consistency all arrows shot.



Variance is how much the arrows hit are spread around the 10 ptr.





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		Overfitting occurs when model capacity is too high bcoz of:
		1) During the training phase, due to the high complexity of the model which captures the noise leading to changes in the prediction of output.
		2) Due to a very large dataset, it tends to read even the unwanted samples & train on it.
		⇒ It affects the generalization bcoz it performs poorly on a new set of data bcoz it is heavily trained on training set leading to vague results.

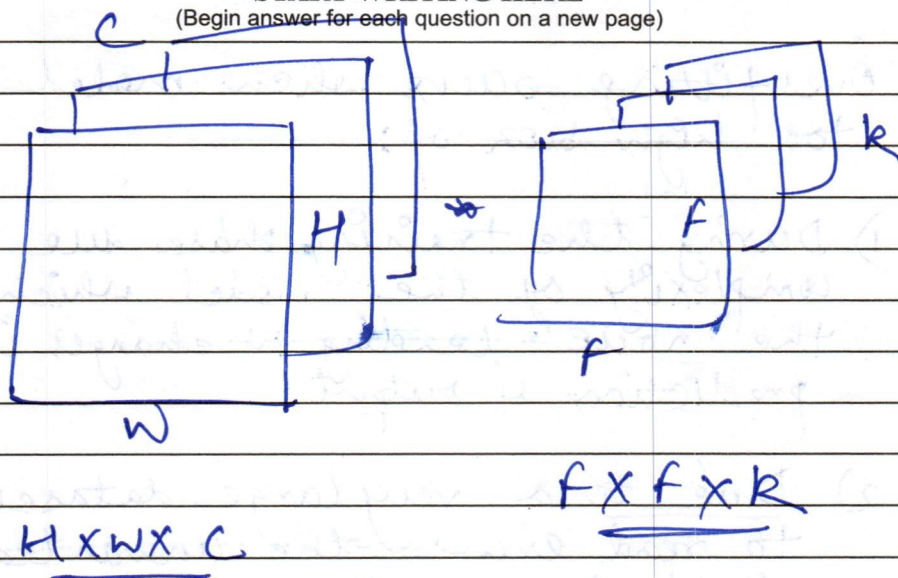
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(Q2)a)

(Q2)a)



Depthwise separable convolutions are critical in modern architectures bcoz:

- ① Captures better spatial relationships from the input image.
- ② Instead of treating every pixel as individual features, it makes sure that all the pixels are convoluted at once bringing out the necessary features representation out of it.

⇒ In terms of parameter efficiency, as it downsamples the input image, the parameters keep on reducing as the large input size dimensions are convoluted & then pooled upon, which ensures better model efficiency & ~~lower~~ ^{better} computational training as parameters which contribute to a better output image are only carried on, in each subsequent layer.

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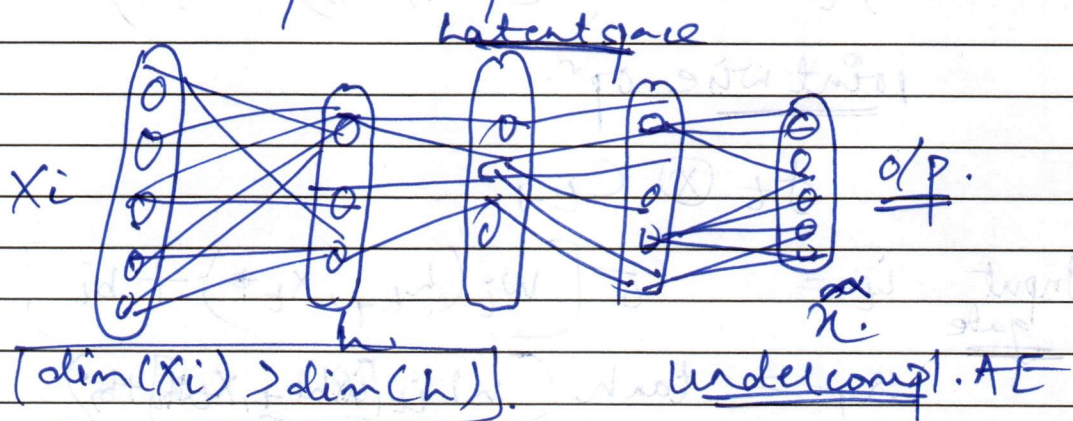
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Q2/b) Linear Undercomplete AE:

Q2/B)

In undercomplete AE, dimensions of x_i is $>$ dimensions of h , leading to selection of feature representation of only those data pts which contribute the most to the output image.

As in PCA, only top 'k' components are selected, in the same way when a linear undercomplete AE is trained with MSE it tends to learn a subspace equivalent to PCA.



⇒ Overcomplete AE tends to learn identity function bcoz:

- 1) input ~~into~~ wt vectors are directly mapped to the latent space which further is directly mapped to o/p, due to which the i/p is almost equal to the o/p bcoz of the linearity of which it was directly mapped.

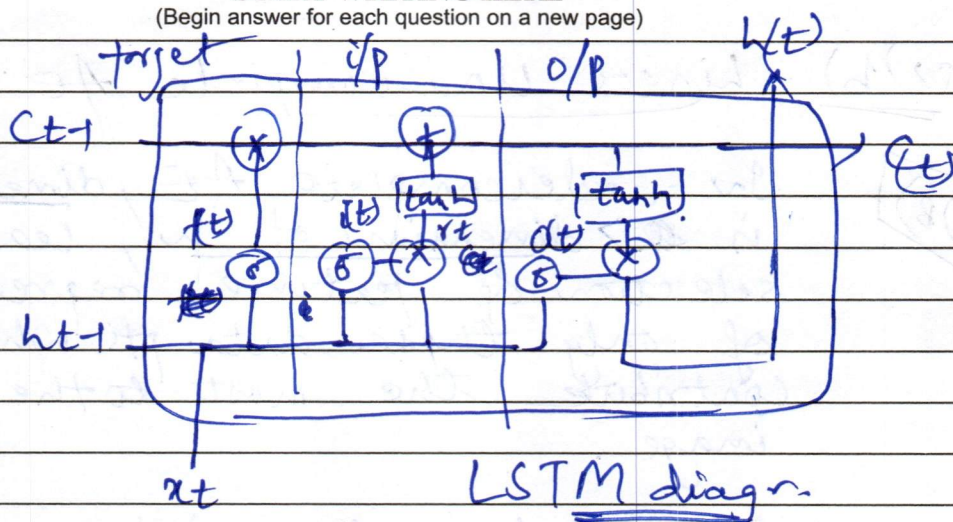
- 2) It doesn't capture any important features as $x_i \approx \hat{x}$

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(Q3A)



$$h_t = o_t \times \tanh(C_t) \Rightarrow \text{LSTM o/p eq.}$$

forget gate

$$f_t = \sigma(W_f(h_{t-1}, x_t) + b_f)$$

point wise opr.

$$f_t \otimes C_{t-1}$$

Input gate

$$i_t = \sigma(W_i(h_{t-1}, x_t) + b_i)$$

$$r_t = \tanh(W_{ir}(h_{t-1}, x_t) + b_{ir})$$

$$C_t = f_t \otimes C_{t-1} \oplus i_t \otimes r_t$$

o/p

$$h_t = o_t \otimes \tanh(C_t)$$

$$o_t = \sigma(W_o(h_{t-1}, x_t) + b_o)$$

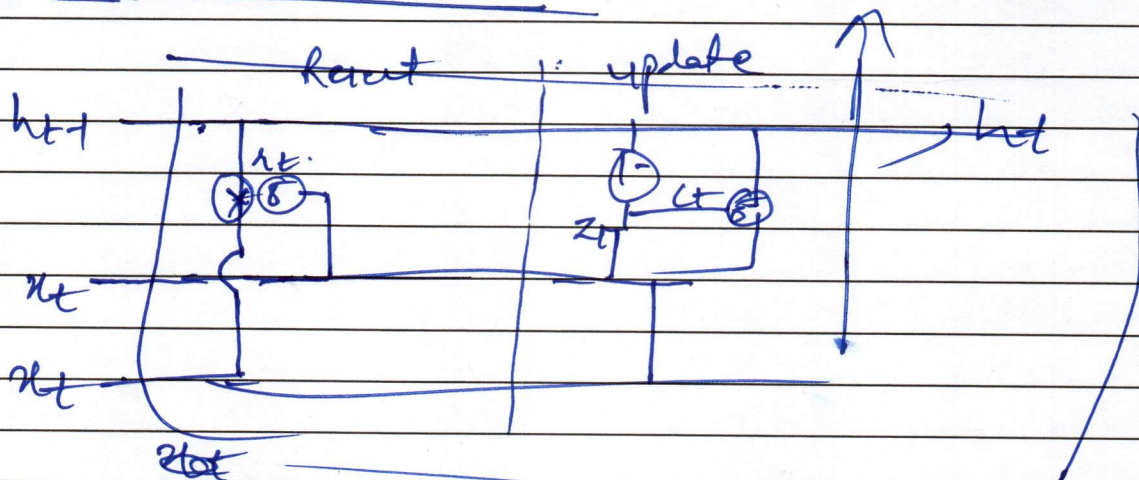
→ forget gate takes into consideration of output from previous cell states as well due to which it ensures that along with C_{t-1}, h_{t-1}, x_t as input the gradients never tend to vanish, capturing long term dependencies along with short terms.

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Q3(B)

GRU (Gated Recurrent Unit)

$$h_t = (1 - z_t) h_{t-1} + z_t \times \tilde{h}_t \quad \text{o/p eq.}$$

GRU tends to have fewer parameters as it tends to make use of fewer gates compared to LSTM using 3 gates & higher parameters. In real world applⁿ GRU's might perform better with large datasets like MNIST.

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Question
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